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Effectiveness of training actions aimed at improving critical thinking in the face of mis- and disinformation: a systematic review

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The threat of mis- and disinformation has prompted international bodies and researchers to propose numerous solutions to mitigate it, highlighting critical thinking (CT) as a key element to counter this phenomenon. Although evidence supports the notion that instruction can improve the ability to think critically in the face of mis- and disinformation, more analysis is needed to determine the most effective ways to develop it. This systematic review (SR) is presented in response to the need to verify the effectiveness of training actions aimed at adults, regardless of gender, origin or professional profile, to improve CT against mis- and disinformation. Specifically, the PRISMA Statement and a previously published SR protocol providing all the methodological details and steps for prospective decision-making were followed. The literature search covered 12 databases and identified 8041 articles. After several rounds of independent peer review, 17 studies were chosen as the final sample. Most of these studies employed true and quasi-experimental designs with mainly moderate risks of bias. These were instructions aimed at media and informational literacy (MIL) delivered through active learning approaches, which included CT as part of the content. The results revealed an overall improvement in the CT of the participants, who were mainly higher education students. However, not all interventions focused on developing this competence intentionally, nor did they measure it specifically. In sum, only seven studies rigorously met the eligibility criteria, highlighting the incipient nature of this topic. Moreover, the findings' heterogeneity hindered a clear verdict on overall effectiveness. Nevertheless, key elements to consider, such as active learning, diversity of resources, and the explicit integration of CT as a core objective within robust experimental designs, were identified. The success of future interventions could depend on these factors, so further research on approaches that conceptualise and evaluate CT appropriately is needed.

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Introduction

The global threat of mis- and disinformation. Mis- and disinformation, although not a new problem, have evolved into a global crisis of considerable magnitude and severe consequences (Bennett and Livingston 2018; Lazer et al. 2018; Valera-Ordaz et al. 2022). This escalation has been fuelled by a convergence of several factors, including modern digital environments, technological advancements, and turbulent social dynamics (Adams et al. 2023; LaGarde and Hudgins 2018; Sługocki and Sowa 2021). Currently, its growing scope impacts virtually all areas of our lives (World Economic Forum 2024, 2025), presenting an unprecedented challenge not only for citizens but, above all, for democratic structures, journalism, and freedom of expression (Lewandowsky et al. 2017; McKay and Tenove 2021; Monsees 2023; Rodríguez-Pérez 2019)[1].

The surge and widespread use of the term "fake news" around 2016 and 2017 underscored the need for a more accurate designation to capture the complexity of the various forms of false and misleading information (Rodríguez and Pérez 2019; Wardle 2018). In response, Wardle and Derakhshan (2017) introduced the notion of "information disorder", conceived as a metaphorical and conceptual, not a clinical, framework to illustrate the systemic and multifaceted nature of this phenomenon. Based on this perspective, they proposed a widely adopted classification distinguishing between *misinformation* (false information shared without intent to deceive), *disinformation* (false information deliberately created or disseminated to manipulate), and *malinformation* (truthful information used maliciously or taken out of context to cause harm). This approach shifted the academic focus towards analysing the phenomenon as an evolving process rather than a static outcome (Sánchez Duarte and Magallón Rosa 2023). Since then, Wardle and Derakhshan's categories—based on the underlying intention behind the information—have underpinned a wide range of taxonomies considering additional factors, such as facticity levels or dissemination channels. These classifications span from unintentional mistakes to organised manipulation strategies (Botha and Pieterse 2020; Molina et al. 2021; Lazer et al. 2018).

While this framework remains influential, it has also sparked increasing scrutiny regarding its adequacy and the need for a more nuanced and applicable conceptualisation. From a practical perspective, organisations such as the European Commission (EC 2018; n.d.) and the OECD (2024) prioritise disinformation as the overarching term, highlighting harmful intent as the primary concern. In contrast, other influential firms, such as the American Psychological Association (APA) (Ecker et al. 2022), advocate for a broader definition of misinformation, including any false or misleading content, regardless of intent. This divergence fuels academic debate, where contrasting perspectives emerge among authoritative voices. For example, Wardle (2023), co-author of the original taxonomy, has called for moving beyond rigid categorisations. In contrast, Hameleers et al. (2022) propose a clearer conceptual separation between misinformation and disinformation, particularly after incorporating audience perceptions as a relevant factor in their analysis.

All these evolving insights enrich the traditional conceptual framework, offering a deeper understanding of the phenomenon. Therefore, given the absence of a unified perspective and the complexity of the issue, this study deliberately adopts both terms —mis- and disinformation—to ensure conceptual breadth and analytical clarity.

Solutions in the face of mis- and disinformation. Recent years have seen a surge in research addressing the impact of mis- and disinformation (Pérez-Escolar et al. 2023; Roozenbeek et al. 2023). Particularly, a significant focus has been placed on identifying solutions (Cover et al. 2023) and categorising them based on their typology, level of implementation (individual or systemic), and timing (reactive or preventive) (Ecker et al. 2022; Quevedo-Redondo et al. 2022; Roozenbeek et al. 2023; van der Linden and Roozenbeek 2020; van der Linden et al. 2021). A large global study identified up to nine types of interventions (Kozyreva et al. 2024).

Reactive strategies, focused on debunking, operate after the spread of mis- and disinformation and constitute a fundamental praxis in journalistic responsibilities (Herrero-Diz et al. 2024). These include regular fact-checking processes (Chan et al. 2017; Nyhan and Reifler 2012), crowdsourced verification (Panizza et al. 2023), the development of detection algorithms (Monti et al. 2019), and the creation of global initiatives and digital media specialised in information verification, as well as early warnings (Andaluz Antón et al. 2022; Clayton et al. 2020; Ferracioli et al. 2022; Vizoso and Vázquez-Herrero 2019). Furthermore, there are legislative initiatives (Oleart and Bouza 2022; Pamment 2020), such as the regulatory strategy promoted by the European Union (EU), which combines stringent security policies to protect its democracies with more flexible measures that encourage voluntary collaboration, particularly among media outlets (Casero-Ripollés et al. 2023).

On the other hand, based on pre-bunking, preventive strategies aim to build resilience before exposure to false or misleading content (Compton 2013; Jolley and Douglas 2017). These approaches can be grouped into both instructional and psychological interventions: instructional interventions through Media and Information Literacy (MIL) programs (Hameleers 2024), critical thinking (CT) training (Muñiz-Velázquez and Marcos-Vílchez 2023), and practices such as lateral reading (Wineburg et al. 2022); and psychological interventions through techniques like technocognition (introducing systemic friction to prompt reflection) (Lewandowsky et al. 2017), nudging (utilising subtle cues such as accuracy prompts) (Pennycook et al. 2021; Pillai and Facio 2023), and boosting (enhancing motivational and cognitive skills) (Hertwig and Grüne-Yanoff 2017).

Particularly noteworthy in this field is the resurgence of the cognitive inoculation theory proposed by Papageorgis and McGuire (1961), which conceptualises resistance to controversial beliefs and attitudes through the activation of "mental antibodies", drawing a biological analogy to vaccines (Basol et al. 2020). Roozenbeek and van der Linden (2018, 2019, 2020) revitalised this strategy through an active and experiential approach to address mis- and disinformation, most notably through *Bad News*, a free serious game where players adopt roles to learn and apply general disinformation techniques. Inspired by its success, similar initiatives were developed (Basol et al. 2021; Morejón-Llamas 2023; Vivion et al. 2022), acknowledging the often limited effectiveness of reactive measures due to the persistence of false beliefs even after debunking (Nyhan and Reifler 2012, 2015; Sally Chan and Albarracín 2023; Stencel et al. 2023) and the faster spread of mis- and disinformation compared to factual content (Vosoughi et al. 2018). Nevertheless, inoculation strategies also face challenges: their effectiveness remains uncertain and context-dependent (Brashier et al. 2021; Maldita.es 2023; Spampatti et al. 2024), and they risk fostering excessive scepticism that may undermine trust even in legitimate information (Modirrousta-Galian and Higham 2023; Van der Meer 2023).

Recognising the challenges and occasionally contradictory evidence, the precursors of active inoculation strategies have emphasised the need to combine multiple approaches for a more effective response to mis- and disinformation (Lewandowsky and van der Linden 2021; Roozenbeek et al. 2020). This view, supported by various studies (Dan et al. 2021; Lewandowsky et al. 2020; Roozenbeek et al. 2023) and international organisations such as Europol, Interpol and UNICRI (Gradoń et al. 2021), is reinforced by research findings confirming the necessary complementarity of pre-bunking and debunking through sophisticated actions (Arcos et al. 2022; Tay et al. 2021).

In this context, technology has become a fundamental component of the global response against mis- and disinformation, with advances in artificial intelligence (AI) algorithms, social networks, text analysis, and dissemination pattern evaluation enhancing the detection and mitigation of false content (Gravanis et al. 2019; Gupta et al. 2022; Zhang and Ghorbani 2020). However, education remains the cornerstone for building resilience (Tamboer et al. 2020). In particular, MIL, also referred to as digital literacy (Adjin-Tettey 2022; Tinmaz et al. 2022), serves as a key instrument for channelling such endeavours, to the extent that institutions such as IFLA (2011), UNESCO (Grizzle et al. 2021; Muratova et al. 2019; Wilson et al. 2011), the OECD (2021), and the EC (2023) regard it as a basic human right and consistently advocate for its promotion as a means to achieve genuine citizen empowerment.

MIL and CT: conceptual convergence in instruction against mis- and disinformation. From an operational point of view, MIL refers to a set of competencies (knowledge, skills and attitudes) that enable individuals to access, comprehend, critically evaluate, and effectively utilise information from the media and other content providers, as well as to engage with them and the digital and technological environments through which it is disseminated. This definition implies a civic commitment to different forms of communication, as well as understanding of how messages are constructed, by whom, and for what purposes, and consciously managing one's interaction with media according to personal, social, political and cultural goals (Frau-Meigs and Corbu 2024; Grizzle et al. 2021; UNESCO 2019; Wilson et al. 2011). In response to the rise of mis- and disinformation, MIL has proven effective (Adjin-Tettey 2022; Lu et al. 2024; Machete and Turpin 2020; McDougall 2019; Valverde-Berrocoso et al. 2022), yet it remains necessary to expand its teaching and strengthen its impact through a stronger emphasis on CT (Al-Zou'bi 2021; Aybek 2016; Buckingham 2019; Calvo et al. 2020; Hollis 2019).

According to UNESCO, within the framework of MIL, CT is defined as "the ability to examine and analyse information and ideas in order to understand and assess their values and assumptions, rather than taking propositions at face value" (Wilson et al. 2011, p.182). The literature widely considers CT as one of the backbones of MIL, acting as an underlying component that drives and gives meaning to it (Alcolea-Díaz et al. 2020; Andersson 2021). The intersection of these two constructs (CT and MIL) has given rise to a shared conceptual vision (López-González et al. 2023), reflected in the concept of "critical literacy", which captures the natural convergence of these competencies in the digital age and highlights it potential to mitigate the impact of mis- and disinformation (Brisola and Doyle 2019; Higdon 2020). Nevertheless, scholars such as Potter (2022) and Potter (2023) warn of ongoing discrepancies about this term and persistent ambiguities in how CT is defined and applied in MIL studies. A key issue arises when the inherent complexity of CT—higher-order thinking involving cognitive skills and emotional dispositions (Facione 1990)—is often overlooked (Potter 2018), leading to diverse interpretations or even the erroneous assumption that its meaning is universally shared and understood. In this regard, Andersson (2021) cautions that when MIL is grounded in a vague or reductive conception of CT, it may unintentionally reinforce narratives of mistrust and polarisation. Similarly, Boyd (2017, 2018) and Nygren and Guath (2019) demonstrate that overly simplistic conceptions of CT (rooted in a gnoseological logic of doubt) may lead to the blanket rejection of information, mirroring the very dynamics of anti-democratic propaganda.

The tensions outlined above call into question the relationship between the two competencies and the actual extent to which MIL, as currently conceived, coherently integrates the development of CT. Boyd (2018) and Nygren and Guath (2019) warn of the risk of neglecting key aspects of CT, such as self-reflection, or replacing the fostering of curiosity, which is one of its main drivers, with a pervasive sense of doubt. This emerging debate has led experts such as Bennett & Livingston (2018) to stress the importance of clearly defining and refining the challenges MIL is intended to address. Likewise, (Phippen et al. 2021) advocates for a more nuanced and explicitly CT-focused MIL approach to avoid the risk of developing ineffective instructional interventions. In light of these considerations, it is essential to emphasise the study of CT as a central and priority element in the fight against mis- and disinformation, ensuring that its complexity is acknowledged and meaningfully addressed. This perspective forms the foundation of the conceptual framework adopted in this systematic review (SR), which explores the interrelationship between interventions, CT, and mis- and disinformation, specifically positioning CT as an essential pillar in addressing current information challenges (Machete and Turpin 2020; Puig et al. 2023).

CT development against mis- and disinformation. CT is widely recognised as one of the "21st century skills" (Scott 2017), due to its essential role in problem-solving and informed decision-making (Mayer 1992; Puig et al. 2021; Thorndahl and Stentoft 2020). It constitutes a key cognitive resource across multiple contexts, enabling individuals to address everyday challenges with greater discernment and autonomy (Davies 2013; Franco et al. 2014). Currently, the development of CT is also closely intertwined with digital competence, because it facilitates the accurate analysis and evaluation of media outlets and online content (Jiménez-Rojo 2020). Therefore, in response to an information ecosystem saturated with mis- and disinformation—where distinguishing fact from falsehood is increasingly complex (Pilgrim et al. 2019)—the EC (2019), the OECD (2021), and UNESCO (2017) have underscored the global urgency of actively fostering CT.

Moreover, CT involves a metacognitive awareness—the possibility to monitor, reflect on, and regulate one's thinking—, which, when intentionally developed, enhances sound judgment and independent decision-making (Bernal et al. 2018; Goldstein and Calero 2022). This self-regulatory approach supports the notion that CT represents a way of interacting and being in the world: a sustained attitude that fosters responsibility, open-mindedness, and ethical engagement with information and others (Paul and Elder 2019, 2020; Phan 2010). These qualities make critical thinkers less vulnerable to both fallacies and their own cognitive biases (Simonovic et al. 2023) and better equipped to respond autonomously and effectively to the pervasive dilemma of what to do and believe (Hunter 2014).

This understanding of CT is grounded in a broader theoretical and intellectual tradition. Like other forms of thinking, such as analytical, logical, and creative thinking, CT is an inherent human capacity (Villa and Poblete 2007; Wals and Jickling 2002) that encourages questioning of reality and drives transformations at both the personal and societal levels (Lipman 1987; Thorndahl and Stentoft 2020). Since Classical Greece, many scholars have advocated for cultivating CT, recognising its influence in various intellectual, scientific, and social advancements, as well as the commitment it encourages (Bezanilla-Albisua et al. 2018; Campos Arenas 2007; Halpern 2013). However, despite this longstanding interest, the literature reveals an enduring conceptual ambiguity (Abrami et al. 2008; Ennis 2016; Hitchcock 2022), fuelled by diverse philosophical, psychological, and educational perspectives developed throughout the 20th century (Bloom 1956; Ennis 1987; Facione 1990; Halpern 1998; Kennedy et al. 1991; Kurfiss 1988; Lewis and Smith 1993; Lipman 1991; McPeck 1981; Paul and Binker 1990; Scriven and Paul 1987; Siegel 1988, and others).

Among the several proposals, Facione (1990) stands out as the sole contribution that achieved a relative agreement across disciplines experts, who defined CT as an autonomous judgement involving cognitive skills—interpretation, analysis, evaluation, inference, explanation, and self-regulation—combined with affective dispositions such as truth-seeking, open-mindedness, analyticity, systematicity, self-confidence, inquisitiveness, and cognitive maturity. This model has provided a solid foundation for operationalising CT in education, upon which researchers have further demonstrated that CT-focused training enhances individuals' CT skills and dispositions (Abrami et al. 2008, 2015; Çeviker Ay and Orhan 2020; Enríquez Canto et al. 2021; Halpern 2013).

Despite all this, the most effective methods of teaching CT remain unclear (Bellaera et al. 2021; Thomas and Lok 2015; Vincent-Lacrin et al. 2019). The literature presents numerous CT instructional models, which vary according to the instructional approach (López Aymes 2012). The classic framework by Ennis (1989) outlines four broad types: the *general* approach, which teaches CT skills and dispositions independently; the *infusion* approach, which explicitly integrates them into other subjects; the *immersion* approach, which incorporates them implicitly; and the *mixed* approach, which combines elements of all three previous approaches. While the infusion and immersion models are widely supported (Abrami et al. 2008; Behar-Horenstein and Niu 2011; Lombardi et al. 2021; Tiruneh et al. 2014), various empirical studies have repeatedly shown that explicit and structured instruction in CT principles tends to be more effective than implicit methods (El Soufi and See 2019; Ennis 1989; Horn and Veermans 2019; Tiruneh et al. 2014). This seems particularly remarkable when adopting elements of the general approach (Orhan and Çeviker Ay 2023), which suggests promising outcomes for mixed models (Simonovic et al. 2023).

Building on this evidence, experts advocate for systematic, intentional, and sustained CT pedagogy grounded in recognised effective strategies such as active and cooperative learning, aligned assessment practices and a deliberate focus on fostering affective dispositions (Dominguez 2018; Elen et al. 2019; Silva et al. 2021, 2022). However, important gaps persist when CT is applied as a tool in the face of mis- and disinformation. Despite its potential, the literature highlights that there are few studies developed in realistic contexts (Lutzke et al. 2019; Roozenbeek et al. 2023; Horn and Veermans 2019), and that, in many cases, its incorporation occurs indirectly through MIL programmes (Rodríguez Mendoza and Zambrano Montes 2019).

This reality poses a significant epistemic challenge (Gallego Torres 2023), as it confronts the previously announced transformative potential with the inherent ambiguity resulting from the extended convergence between MIL and CT. While MIL primarily focuses on critical engagement with information and media in digital environments (Grizzle et al. 2021), CT brings a broader dimension, i.e. a vital and ethical attitude towards the world, which requires specific approaches for both its development and assessment (Abrami et al. 2015; Dominguez, 2018; Scriven and Paul 1987; Tiruneh et al. 2014). This scenario compels a direct question of the role of CT in addressing mis- and disinformation, which is increasingly understood as a global phenomenon permeating all levels of social life. Consequently, further research is required to clarify how these two competencies can be coherently integrated into educational practice. But before, we emphasise that the relationship between CT and vulnerability to mis- and disinformation warrants close examination to understand its causal mechanisms and strengthen the design of effective interventions (Escolà-Gascón et al. 2021; Herrero-Diz et al. 2023; Olariu 2022).

Current study

Justification for the study. Based on the theoretical framework outlined above, it is evident that the lack of clarity regarding effective methods of developing CT and addressing mis- and disinformation, coupled with an insufficient focus on the critical dimension of literacy efforts, may limit students' training to engage with information and assess its credibility critically (Phippen et al. 2021; Weiss et al. 2020). Given the growing need for appropriate interventions to counter the escalating challenge of mis- and disinformation (Hobbs 2022; Saltz et al. 2021), there is an urgent need for a literature review to identify initiatives that promote CT as an essential competence to combat this problem, and to shed light on how to overcome the conceptual ambiguities observed, where CT is often assumed rather than explicitly taught or assessed. Therefore, the present SR does not aim to evaluate all types of training to address the challenges of mis- and disinformation per se, but to specifically examine those focused on CT development.

To date, to the best of our knowledge, only the SR by Machete and Turpin (2020) has directly addressed the role of CT in identifying fake news, albeit with certain limitations. Consequently, the authors agreed on the need for a new review to provide clearer insights for future research efforts. Under this premise, the present SR was conducted to bridge existing gaps and respond to emerging research demands, following the methodological criteria recommended by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020 Statement) (Page et al. 2021) and a preceding SR protocol article (Marcos-Vílchez et al. 2024).

Research objectives. The main objective[2] of this SR is to examine the effectiveness of training actions, especially those with an instructional intervention design aimed at adults (of legal age), regardless of their gender, origin or professional profile, to improve CT in the face of mis- and disinformation.

To that end, three specific research objectives have been established:

1. To identify and compile available evidence on training actions aimed at improving CT in the face of mis- and disinformation.
2. To systematise the key elements of these training actions.
3. To evaluate the results of these training actions.

Methods

SR design and protocol. The results of this SR were reported in accordance with the PRISMA 2020 Statement (Page et al. 2021), whose checklist of 27 numbered items was meticulously completed and used to structure the different sections of the manuscript (see Supplementary Appendix II). The four authors involved [(JMMV), (JAMV), (AAC), (M.S.M)] collaborated throughout the review process and worked together.

In addition, as mentioned above, an SR protocol was previously designed and published in a scientific journal after rigorous peer review (Marcos-Vílchez et al. 2024). This paper ensured that the methodological decisions of this SR were prospectively traced,

Table 1 Inclusion and exclusion criteria.

Eligibility criteria

Criteria	*Inclusion criteria*	*Exclusion criteria*
Participants/population	Adults (of legal age), regardless of gender, origin and professional profile.	Other participants, mainly minors.
Intervention	Studies in which training actions are developed through an instructional or psychological intervention[a] design to improve CT in the face of mis- and disinformation.	Other types of interventions.
Comparators	Control groups without training. Training actions not specifically aimed at improving CT in the face of mis- and disinformation. Other types of instructional strategies. Interventions without necessarily educational or training purposes.	Other comparators.
Outcomes	The improvement of CT skills and/or dispositions in the face of mis- and disinformation as an outcome measure to assess the effectiveness of training actions. These outcome variables should be measured through validated or non-validated CT measuring instruments.	Other results.
Study design	Experimental or quasi-experimental primary studies will be selected, although other pre-experimental designs could also be considered.	Pre-experimental design studies without pre-post measures. Other empirical non-experimental or observational studies. Secondary research studies include literature reviews (systematised and non-systematised) and meta-analyses.
Types of scientific publications	Scientific articles on empirical studies published in research journals, indexed in one of these 12 electronic databases consulted: Web of Science (WOS), SCOPUS, Education Resource Information Center (ERIC), Communication Source, PsycInfo, Psychology and Behavioral Sciences Collection, Psycarticles, Medline, Education Source, Communication Source, Academic Search Ultimate and Humanities Source Ultimate.	Books, book chapters, conference or congress papers, doctoral theses and dissertations. Other non-scientific or informative publications.
Languages	All languages.	None.
Settings	All settings.	None.
Time period	Studies of any date given the breadth we seek in the review.	None.

Source: Authors.
[a]Although the PICO strategy developed for the original protocol focused on instructional interventions, the corresponding eligibility criterion (type of intervention) was subsequently broadened to include psychological interventions explicitly aimed at fostering CT in response to mis- and disinformation. This decision followed a deeper engagement with the theoretical framework of the SR, which clarified that both instructional and psychological approaches can be meaningfully operationalised as formative actions. The adjustment reflects the practical convergence of both types of interventions within this field and is documented in Section Differences between the protocol and the SR.

verified and validated beforehand, complying with the standards established by PRISMA 2020 (Page et al. 2021), the Cochrane Collaboration (Higgins et al. 2024) and the Campbell Collaboration (www.campbellcollaboration.org). Similarly, this protocol study was also reported following the PRISMA-P 2015 Statement (Moher et al. 2015; Shamseer et al. 2015).

Eligibility criteria. The inclusion and exclusion criteria for the selected studies were defined on the basis of the PICOS model (Higgins et al. 2024; Martínez Díaz et al. 2016) and additional items—types of scientific publications, languages, settings and time frames—to develop a comprehensive assessment of studies that present training actions aimed at adults (legal age), regardless of gender or professional profile, to improve CT in the face of mis- and disinformation (see Table 1). Detailed explanations of each of these elements were outlined in the SR protocol (Marcos-Vílchez et al. 2024).

Search strategy and study selection process. The bibliographic search was conducted entirely by JMMV It began in October 2022 and was updated in November 2023, according to the corrected version of the protocol, across 12 electronic databases: Web of Science (WOS), SCOPUS, Education Resource Information Center (ERIC), Communication Source, PsycInfo, Psychology and Behavioral Sciences Collection, PsycArticles, Medline, Education Source, Communication Source, Academic Search Ultimate, and Humanities Source Ultimate. The search term string, designed according to the structure of the PICOS model, was initially tested on WOS and SCOPUS (see Supplementary Table A1 in Supplementary Appendix III) and then adjusted and replicated across the remaining databases.

To ensure conceptual and methodological clarity, and alignment with the objectives of this review, the search and selection strategy focused exclusively on studies explicitly referencing “critical thinking” or closely related terms such as “critical reasoning” or “critical judgement”, as defined in the syntax used in the referenced search string. This approach was applied to the full text of publications and did not include broader or substitute conceptual indicators, nor was an alternative search strategy implemented.

Although the construct of CT is subject to ongoing disciplinary debate, the term itself is firmly established in academic discourse and widely recognised as the standard label used to denote its epistemological domain (Davies 2013; Elder 2022; Ennis 2015). Therefore, privileging this terminology ensures alignment with the prevailing discourse of the field and supports a systematic and comparable synthesis of the available evidence. While this decision may have excluded studies that implicitly foster CT-related skills under other labels, it strengthens the methodological coherence, replicability and conceptual consistency recommended by the PRISMA 2020 Statement (Page et al. 2021) and the Cochrane Handbook (Higgins et al. 2024; Lefebvre et al. 2025), as well as the overall scope of this review.

A total of 9830 studies were retrieved, downloaded in RIS (Research Information Systems) format, and imported into the Rayyan (https://www.rayyan.ai/), an online platform supporting blind and independent screening throughout the process (Ouzzani et al. 2016). These studies were grouped into a single database and cleaned of duplicates, incomplete, and unreadable files by JMMV, resulting in the removal of 1789 studies. Subsequently, 8041 entries were 'masked' and independently reviewed by JMMV and AAC, who filtered them according to the eligibility criteria by reading titles and abstracts starting in December 2023. This task achieved a 95% agreement rate, but it was necessary to pool disagreements to reach a 100% consensus. A total of 7898 entries were excluded because they did not meet the eligibility criteria. The main reasons for exclusion, listed in order of frequency, are incompatibilities in the following areas: background article (49.86%), outcomes (24.86%), publication type (13.03%), study design (10.24%), and population (2.01%).

After this initial filter, 143 studies were selected for full-text review. In this phase, the authors involved, JMMV and JAMV, undertook a blinded peer review in which they identified publications with the best fit to the eligibility criteria before starting data extraction and analysis. This second screening resulted in substantial agreement among the authors (kappa = 0.71, 95% CI 0.61–0.83), according to the scale proposed by Landis and Koch (1977). Disagreements were discussed until a consensus was reached on the final number of studies included in the SR.

Due to the limited number of publications reporting directly measured improvements in CT, the authors agreed to apply a degree of controlled tolerance regarding the eligibility criterion related to expected outcomes, as defined by the PICOS model. Accordingly, in addition to studies reporting specific measured results, we also included those that, while not providing direct empirical assessments, provided well-reasoned inferential analyses linking the intervention to changes in CT. We ensured that these studies met all remaining eligibility criteria—including target population, methodological design, and, crucially, an explicit focus on CT development—thereby guaranteeing full alignment with the objectives of this SR.

This strategy is supported by leading methodological guidelines for SRs (Booth et al. 2021; Gough et al. 2017; Higgins et al. 2024), particularly in complex or emerging fields, where controlled adjustments to inclusion criteria are deemed acceptable, provided such decisions are informed, non-arbitrary, and well justified. Moreover, such modifications are especially plausible when they enhance the interpretative value of the synthesis and enable a more comprehensive picture of the available evidence without compromising methodological rigour. Thus, this approach allowed us to broaden the scope of included studies in the SR, provided the intention to promote CT was clearly articulated, while also deepening the interpretation of findings by distinguishing between studies reporting directly measured outcomes and those offering inferential or indirect assessments. To ensure analytical transparency and procedural traceability, this protocol deviation has been consciously implemented, justified by relevant literature and documented in alignment with PRISMA 2020 recommendations (Page et al. 2021), as detailed in Section "Differences between the protocol and the SR". Differences between the protocol and the SR.

A final sample of 17 studies was selected, while 126 entries were excluded due to incompatibilities with eligibility criteria related to an unsuitable study design, outcomes, population, and publication type. Among these exclusions were several influential contributions to the field of countering mis- and disinformation, such as those by Cook et al. (2017, 2018, 2022), Puig et al. (2021), and Roozenbeek et al. (2023). These studies were not included because they did not present an explicit and intentional focus on the development of CT, as defined in our eligibility criteria. Although labelling variations may have contributed to these and other paper exclusions, consistency was maintained in using a widely recognised term, 'critical thinking', that serves as a stable identifier of its epistemological domain, as previously discussed. In any case, the lack of conceptual consensus surrounding CT is effectively acknowledged, not a methodological concern in this SR, but reflects a structural challenge within the literature. This issue has been noted in the conclusions as a general limitation and a promising avenue for future research. The complete study acquisition process is visually represented in Fig. 1, via the PRISMA 2020 flowchart (Page et al. 2021).

Data collection and synthesis. Once the studies were selected, JMMV and AAC extracted the most relevant data for this review. Then, the analysis of the findings was carried out descriptively through a systematic process undertaken by JMMV

Quality assessment. To determine the quality of evidence, the authors assessed the risk of bias in individual studies using two assessment tools: ROB-2.0 (Risk of Bias 2.0, Sterne et al. 2019) and ROBINS-I (Risk of Bias in Non-randomised Studies of Interventions, Sterne et al. 2016).

Results

The 17 studies included in this SR, distinguishing between those that explicitly assessed CT ($n = 7$) and those that did so indirectly ($n = 10$), were analysed in line with the predefined specific research objectives. The main findings are presented below.

General characteristics of the selected studies (research objective 1). As observed in the table summarising the general evidence (see Table 2), the 17 studies selected were published between 2017 and 2023, with the highest output in 2022 ($n = 7$). All the papers were written in English and published in high-impact journals by academics from different countries, often in inter-institutional collaboration.

The total number of participants in the interventions analysed was 4972 individuals, mainly university students and predominantly women (see Fig. 2), from various disciplines, including communication (Cernicova-Buca and Ciurel 2022; Schilder and Redmond 2019; Şencan and Soydal 2023), psychology (McLaughlin and McGill 2017; Motz et al. 2022; Simonovic et al. 2022), health sciences (Crist et al. 2017), history (McLaughlin and McGill 2017), and education (Tulis 2022). In the remaining cases, the professional profiles were unspecified (Ahmadi et al. 2021; Green et al. 2022; Lutzke et al. 2019; Murrok et al. 2018 this one is: Murrock et al. 2018; Nazarnia et al. 2023; Yang et al. 2020) or involved very specific samples, such as secondary school teachers (Caroti et al. 2022) or personal trainers (Jolley et al. 2020).

In terms of design, the studies were evenly split between true experimental and quasi-experimental approaches ($n = 12$, and 6 respectively). In the true experiments, all the studies adopted a between-subjects design, mainly with two or three randomly assigned groups of participants. In the quasi-experiments, only one study (time series data) adopted a within-subjects design with one group of randomly assigned participants (Al-Zou'bi 2022), and the rest adopted a between-subjects design with two groups of participants non-randomly assigned. The other five studies adhered to a mixed-methods experimental design, combining

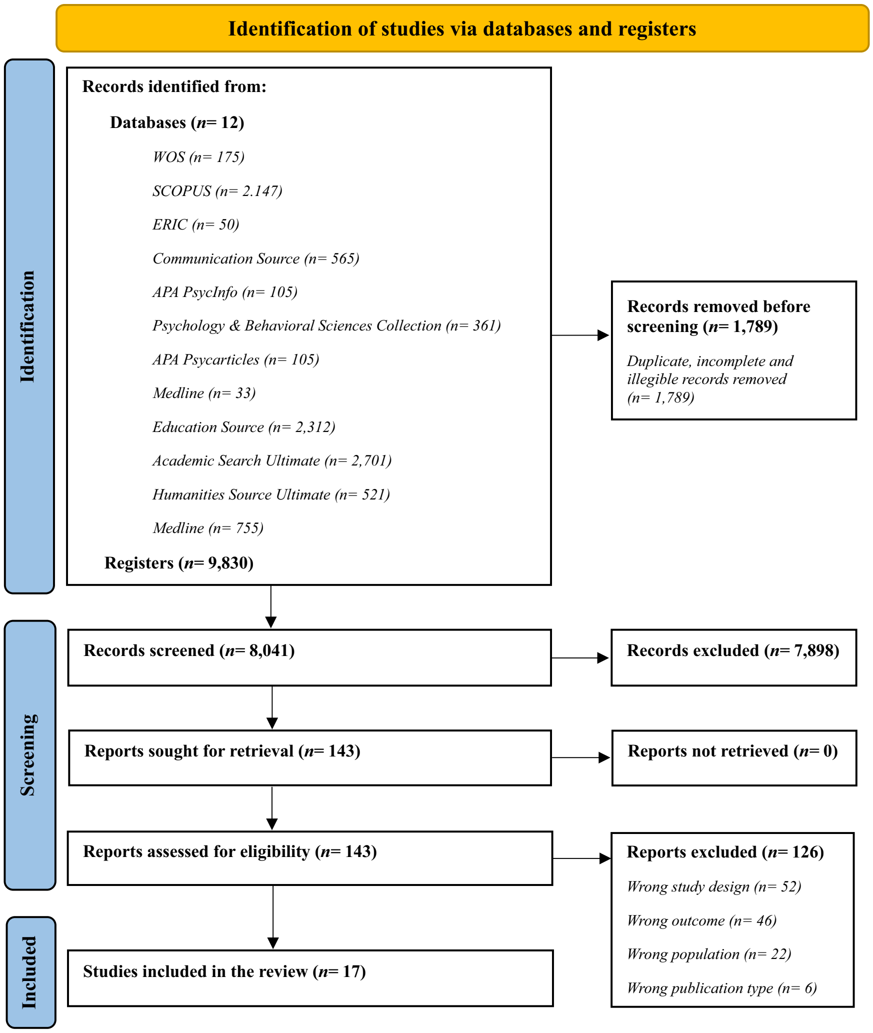

Fig. 1 PRISMA 2020 flow diagram for a systematic review, which included searches of databases and registers only (Haddaway 2022 et al.; Page et al. 2021). Source: Authors.

between-subjects and within-subjects experimental conditions with three or two randomly and non-randomly assigned groups (Green et al. 2022; McLaughlin and McGill 2017; Simonovic et al. 2022), as well as a pre-experimental design with a single group tested pre- and post-intervention (Cernicova-Buca and Ciurel 2022; Schilder and Redmond 2019).

Most studies ($n = 13$) expressly pursued CT promotion and improvement in the fight against mis- and disinformation, often integrating this goal with broader objectives. These objectives included tackling pseudoscience or fallacies in general (Caroti et al. 2022; Motz et al. 2022), determining the impact of MIL competencies (Ahmadi et al. 2021; Al-Zou'bi 2022; Crist et al. 2017; Murrock et al. 2018; Schilder and Redmond 2019; Simonovic et al. 2022), and other, more specific aims, such as reducing unjustified beliefs (UBs) in the field of history (McLaughlin and McGill 2017), and misconceptions about psychology (Tulis 2022), the sports training environment (Jolley et al. 2020) or climate science and change (Green et al. 2022; Lutzke et al. 2019).

The remaining four studies did not explicitly target CT improvement, still it was aimed expressly through specific related purposes, such as testing the value of serious games as a pathway to greater resistance to mis- and disinformation (Cernicova-Buca and Ciurel 2022), assessing the relevance of fostering skills such as scepticism and information discernment, especially in digital environments (Yang et al. 2020), determining the effects of an educational intervention based on a mobile application on the development of media health literacy (MHL) (Nazarnia et al. 2023) and examining the benefits of fact-checking training modules within the verification process (Şencan and Soydal 2023).

Training actions analysis (research objective 2). The analysis of the key elements of the training actions focused on their type and design, thematic focus (domain), and delivery methods (see Supplementary Table A2 in Supplementary Appendix III).

All the interventions analysed met the eligibility criteria: most of them adopted an instructional approach ($n = 15$), while two employed psychological strategies, such as cognitive inoculation, which align more closely with behavioural approaches (Cernicova-Buca and Ciurel 2022; Green et al. 2022). Regardless, all these interventions were categorised into three main types: ad hoc interventions specifically designed for experimental purposes ($n = 8$), those embedded within existing curricula or courses ($n = 7$), and two initiatives designed according to prior models as adapted replications (Caroti et al. 2022; Crist et al. 2017). While many actions were implemented as complete stand-alone courses ($n = 5$) or were integrated as sessions or parts of training in broader programmes ($n = 4$), others featured different educational formats, such as online games (Yang et al. 2020), digital

Table 2 General characteristics of the selected studies.

Interventions that explicitly measured CT

	Study identification and authorship		*Participants*					*Study design*		*Study purpose*
No	**Author and year**	**Institution and study region**	**Sample (*n*)**	**Mean age (SD)/ Interval**	**Gender (F)**	**Gender (M)**	**Professional profile**	**Type of experimental design**	**Grps**	**Research objective**
1	Caroti et al. (2022)	-Aix Marseille University (France) -Keele University (UK)	130 NR	44.6	69.3% *90*[a]	30.7% *40*[a]	Teachers (Secondary)	Quasi-experiment (Between-subjects)	2 nr	Evaluate the effectiveness of an intervention designed to promote CT and educate participants on the psychological difference between science and pseudoscience.
2	Jolley et al. (2020)	-Curtin University (Australia) -Edith Cowan University (Australia)	180R	-Intervention: 41.69 (SD = 11.75) -Control: 39.45 (SD = 11.79)	36 52 *88*	9 28 *37*	Personal trainers	True experiment (Between-subjects)	2 r	Assess the impact of an online domain-specific CT intervention on the presence of misconceptions, knowledge, and CT skills of personal trainers.
3	Motz et al. (2022)	Indiana University Bloomington (USA)	Study 1: 360 NR Study 2: 253R	Study 1: 19.9 Study 2: ≥18 years	50.2% *180*[a] ND	49.8% *180*[a] ND	HE students ≤18 years old in USA	Study 1: Piloting Study 2: True experiment (Between-subjects)	– 3 r	To examine whether an inductive categorisation training regime improves overall CT performance, where students must identify fallacies and biases in the presented statements and classify them.
4	Ahmadi et al. (2021)	University of Medical Sciences, Yaz (Iran)	88R	19–22 = 22 23–26 = 46 27–30 = 20	56	32	HS students BSc = 27 MSc = 22 Ph.D. = 39	True experiment (between-subjects)	2 r	Determine the effect of educational programs on the ML of medical students of Shahid Sadoughi in Yazd in 2020, where CT skill is integrated.
5	Simonovic et al. (2022)	University of Derby (UK)	58R	28.01 (SD = 8.62)	62.2% *36*[a]	37.7% *22*[a]	HE students	Mixed design experiment (Experimental conditions between- and within-subjects)	2 r	Examine the effect of online students' perceptions and attitudes towards CT on confidence, appraisal, misconceptions, cognitive reflection, and authors' writing.
6	Crist et al. (2017)	-Mississippi State University (USA) -Virginia Polytechnic Inst. and State University (USA)	59 NR	ND	ND	ND	HE students	Quasi-experiment - Cohort study (Between-subjects)	2 nr	Improve students' IL skills, including obtaining and discerning relevant and reliable scientific information, developing their CT, and using scientific language (communication skills) after completing a Wiki research project and interdisciplinary conferences in a functional food university course.
7	Tulis (2022)	University of Salzburg (Austria)	184 NR	20.24 (SD = 2.4)	85% *157*[a]	15% *27*[a]	HE students	Experimental field study pre-post-follow-up design (Between-subjects)	2 r	To investigate a simple intervention that can reduce psychological misconceptions among prospective teachers enrolled in an introductory psychology course, leading to increased CT and evaluativist epistemic beliefs and decreased absolutist epistemic beliefs.

Interventions that indirectly assessed CT

	Study identification and authorship		*Participants*					*Study design*		*Study purpose*
No	**Author and year**	**Institution and study region**	**Sample (*n*)**	**Mean age (SD)/ interval**	**Gender**		**Professional profile**	**Type of experimental design**	**Grps**	**Research objective**
8	Al-Zou'bi (2022)	Al al Bayt University (Jordan)	100R	ND	81	19	HE students	Quasi-experiment with time-series data (Within-subjects)	1 r	Assess the impact of MIL, where CT is integrated, on undergraduate students in enabling them to detect fake news (disinformation).
9	Schilder and Redmond (2019)	-RIPE NCC (The Netherlands) -Appalachian State University (USA)	72 NR	(18–21)	ND	ND	HE students	Pre-experimental design with pre- and post-intervention testing	1 nr	To examine the impact on complex and CT skills by evaluating questions that participants ask about media news before and after an ML learning course.
10	Murrock et al. (2018)	International Research and Exchanges Board - IREX (USA and Ukraine)	412R	18–30 = 39% 31–55 = 42% ≥56 = 19%	BS	BS	66% HE level	Cross-sectional quasi-experiment (Between-subjects)	2 nr	Assess the long-term impact of a Learn to Discern (L2D) ML training course by evaluating participants' and non-participants' ML skills.

Table 2 (continued)

Interventions that indirectly assessed CT

	Study identification and authorship		*Participants*					*Study design*		*Study purpose*
No	**Author and year**	**Institution and study region**	**Sample (*n*)**	**Mean age (SD)/ interval**	**Gender**		**Professional profile**	**Type of experimental design**	**Grps**	**Research objective**
										The course's purpose is to teach citizens to identify mis- and disinformation in the news media and foster CT.
11	McLaughlin and McGill (2017)	North Carolina State University (USA)	119 NR	-History. Frauds and Mysteries: 20.24 (SD = 2.20) -History. Honours seminar of the previous course: 19.9 (SD = 0.9) -Psychology. Research Methods: 20.14 (SD = 2.92)	78	41	HE students	Mixed design experiment (Experimental conditions between- and within-subjects)	3 nr	To investigate the effectiveness of explicit instruction in CT skills to reduce unjustified beliefs (UBs) through a history course and an honours seminar focused on exposing "pseudohistory", "pseudoscience", and "pseudoarchaeology" compared with a research methods course in psychology.
12	Lutzke et al. (2019)	-University of Michigan (USA) -Decision Science Research Inst. (USA)	2750R	44.11 (SD = 15.10)	BS	BS	44% HE level	True experiment (3×2 between-subjects design)	2 r	Examine the impact of priming CT through reading or interacting with guidelines on assessing the credibility of news posts on Facebook on people's trust, liking, and sharing of real and fake news about climate change.
13	Cernicova-Buca and Ciurel (2022)	Politehnica University Timișoara (Romania)	50 NR	21.9	ND	ND	HE students	Pre-experimental design with pre- and post-intervention testing	1 nr	Evaluate whether a serious-game intervention (CRAAP-based, role-play) enhances students' critical appraisal and resilience to mis- and disinformation, and explore their perceptions of its educational value as future communicators.
14	Green et al. (2022)	James Cook University (Australia)	137 NR	36.63 (SD = 13.01)	88	49	ND	Mixed design experiment (3 experimental conditions)	2 r	Compare the effectiveness of active versus passive inoculation in reducing susceptibility to mis- and disinformation in climate science (e.g., persuasion and perceived reliability), assuming an improvement in CT skills.
15	Yang et al. (2020)	Seoul National University (South Korea)	210R	26	105	105	ND	True experiment (Between-subjects)	3 r	Examine the effects of the online educational game on scepticism towards online information and information discernment skills, two of the leading MIL competencies.
16	Nazarnia et al. (2023)	-Tarbiat Modares University (Iran) -Arak Univ. of Medical Sciences (Iran)	100 NR	-Intervention: 37.72 (SD = 6.98) -Control: 35.02 (SD = 8.96)	44 40 *84*[a]	6 10 *16*[a]	Unemployed 38 Employee 49 Freelancer 7 Retired 9	Quasi-experiment (Between-subjects)	2 r	This study aimed to determine the effects of a mobile app-based educational intervention on media health literacy (MHL) development among Iranian adults.
17	Şencan and Soydal (2023)	Hacettepe University (Turkey)	70 NR	ND	ND	ND	HE students	Quasi-experiment (Between-subjects)	2 nr	To assess the effect of the NL training module on the fact-checking behaviour of the students, which aims at a higher critical ability.

Source: Authors.
BS Balanced sample, *BSc* Bachelor's degree, *F* Female, *Grps* Groups/conditions, *HE* Higher Education, *HS* Health Science, *IL* Information literacy, *M* Male, *ML* Media literacy, *MSc* Master of Science, *ND* No data/ Unspecified, *NL* News Literacy, *No* Study identification number, *nr* Non-random assignment, *NR* Non-random sampling, *Ph.D.* Doctorate, *r* Random assignment, *R* Random sampling, *SD* Standard deviation.
[a]Authors' estimate/calculation.

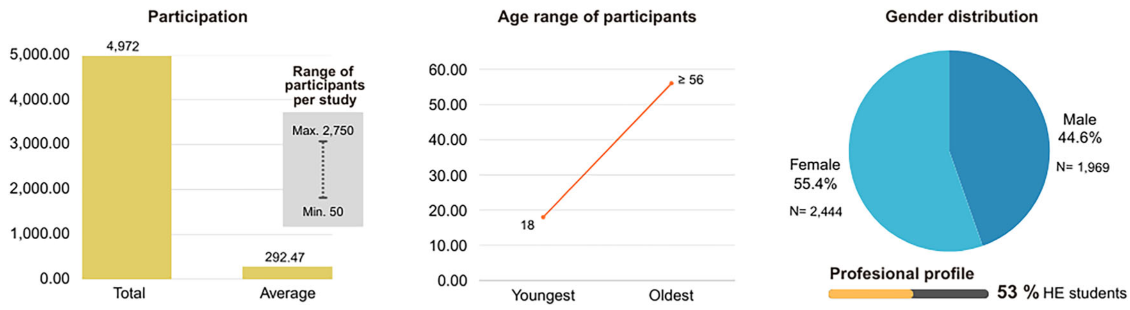

Fig. 2 General characteristics of the sample in the set of interventions analysed. Source: Authors.

apps (Nazarnia et al. 2023) and multimodal approaches (Cernicova-Buca and Ciurel 2022).

With respect to the thematic focus, all the studies addressed CT somehow, but only one study (Simonovic et al. 2022) did so based on a general domain (not applied to any subject). Four interventions treated CT development as a core objective but applied it to specific contexts, such as the refugee crisis (Cernicova-Buca and Ciurel 2022), in response to mis- and disinformation and pseudoscience (Caroti et al. 2022), for personal trainers (Jolley et al. 2020), or aimed at introductory psychology training (Motz et al. 2022). The remaining studies centred on MIL as the primary focus (Al-Zou'bi 2022; Murrock et al. 2018; Yang et al. 2020) but aligned with fields such as education and communication (Schilder and Redmond 2019), health (MHL) (Ahmadi et al. 2021; Nazarnia et al. 2023), and climate change disinformation (Green et al. 2022; Lutzke et al. 2019).

Active, student-centred learning approaches dominated the delivery methods of the training actions ($n = 10$), often leveraging technology to foster interactive learning environments. Common techniques included game-based learning (Cernicova-Buca and Ciurel 2022; Yang et al. 2020), problem-solving and case analysis (Al-Zou'bi 2022; Lutzke et al. 2019), facilitated discussions and debate (Jolley et al. 2020) and project-based activities or expert mediation of training sessions (Crist et al. 2017). Conversely, a smaller subset of interventions ($n = 5$) combined active and passive strategies, blending traditional lectures with collaborative activities and real-world case studies (Crist et al. 2017). Resource use was equally diverse, ranging from written materials and textbooks to multimedia tools and applications, underscoring the flexibility of these interventions. Finally, the duration of interventions varied significantly, with short-term initiatives lasting from a few minutes to a few sessions ($n = 7$) and long-term programmes spanning several weeks to nine months ($n = 8$).

Evaluation of the results of interventions (research objective 3). The evaluation of the results of the interventions considered measurement processes, data analysis, and the findings reported by the individual publications (see Supplementary Tables A3 and A4 in Supplementary Appendix III).

Most studies incorporated pre- and post-intervention assessments to determine their effectiveness, with only minimal exceptions: one study included only a post-test (Lutzke et al. 2019), while three implemented multiple follow-up evaluations to monitor the durability of the effects over time (Al-Zou'bi 2022; Nazarnia et al. 2023; Tulis 2022). Data collection was primarily based on online questionnaires, designed ad hoc for each study, mainly drawing on the scientific literature. Among the most used statistical techniques for data analysis were *t*-tests ($n = 4$) and analyses of variance (ANOVAS) ($n = 6$), which some studies ($n = 5$) also complemented with an analysis of covariance (ANCOVA) and regression models.

Virtually all the interventions achieved their research objectives and had significant positive effects on most study variables. However, notable disparities were observed in how outcomes were measured, explained by the mentioned split between interventions that explicitly evaluated CT ($n = 7$) and those that did not ($n = 10$). While this fact represents a conclusive reality for our SR, this discrepancy also presented challenges for the analysis and direct comparison of findings; for this reason, we explain the results separately for each group (see Supplementary Table A5 in Supplementary Appendix III).

- Interventions that explicitly assessed CT: Studies that directly and explicitly assessed CT as a measurable component provided clearer and more precise evidence for CT improvement in the face of disinformation. They used multiple standardised tools, including reasoning tests, attitude questionnaires and performance assessment rubrics. When adapted to specific contexts and applied in combination, these tools were able to address different dimensions of CT: (1) cognitive skills, measured through the different versions of the Cognitive Reflection Test (CRT) (Caroti et al. 2022; Jolley et al. 2020; Simonovic et al. 2022) and the modified Psychology Critical Thinking Exam (PCTE) (Motz et al. 2022); (2) attitudes and beliefs, related to dispositions to think critically and assessed through instruments such as the Critical Thinking Toolkit (CriTT) (Simonovic et al. 2022) and the Motivated Strategies for Learning Questionnaire (MSLQ) subscale combined with a Misconceptions Scale (Tulis 2022); (3) performance on tasks requiring CT, weighted through an evidence-based rubric designed for use in a specific educational environment and a project work (Crist et al. 2017); and (4) Media literacy (ML) competency explicitly linked to CT, assessed through Zolfaghari's Standard Media Literacy Questionnaire (Ahmadi et al. 2021).

  With respect to reported effectiveness, these interventions demonstrated the potential of CT-focused training to address mis- and disinformation, although the results varied across different contexts. For example, CT improvements that targeted specific skills were observed, such as a reduction in UBs, particularly conspiratorial and supernatural beliefs (Caroti et al. 2022), the enhancement of ML (Ahmadi et al. 2021), the fostering of analytical skills through categorisation tasks (Motz et al. 2022) and the strengthening of scientific evaluation skills through an interdisciplinary project (Crist et al. 2017), which highlighted the effectiveness of active learning designs. However, despite these successes, some shortcomings were also identified. One study reported limited differentiation between the experimental and control groups (Jolley et al. 2020), while another noted a lack of impact on specific

Table 3 Summary of findings on developing, measuring, and improving CT in interventions.

No	*Orientation towards CT improvement*	*Integration of CT within the content (Thematic domain)*	*Instruction approach* (Ennis 1989)	*Explicit CT measurement (Instrument)*	*Measured CT improvement*
Interventions that explicitly measured CT					
1	✓	✓✓ (Specific domain on CT)	Infusion	✓ (CRT, Frederick 2005)	✓
2	✓	✓✓ (Specific domain on CT)	Mixed	✓ (CRT, Primi et al. 2016)	✓
3	✓	✓✓ (Specific domain on CT)	Infusion	✓ (PCT, Lawson et al. 2015)	✓
4	✓	✓	_	✓ (ML Questionnaire, Zolfaghari and Ahmadi 2016)	✓
5	✓	✓✓ (General domain on CT)	General	✓ (CriTT, Stupple et al. 2017)	✓
6	✓	✓	_	✓ A rubric designed by the authors based on the literature.	✓
7	✓	✓	_	✓ (MSLQ, Pintrich et al. 1993)	No
Interventions that indirectly assessed CT					
8	✓	✓	_	No	_
9	✓	✓	_	No	_
10	✓	✓	_	No	_
11	✓	✓	_	No	_
12	✓	✓	_	No	_
13	✓	✓	_	No	_
14	No	✓	_	No	_
15	No	✓	_	✓ Through an item in self-reported pretest questionnaires based on the intellectual civic skills index (Miller 2004; Patrick 2022).	No
16	No	✓	_	No	_
17	✓	✓	_	No	_

No: Study identification number; ✓: CT is addressed; ✓✓: CT is the main topic. Source: Authors.

attitudes or beliefs (Simonovic et al. 2022). Additionally, mixed results were observed for interventions targeting epistemic beliefs (Caroti et al. 2022) or misconceptions (Tulis 2022), raising questions about whether CT training alone is sufficient to effectively counter mis- and disinformation and the possibility of complementing these interventions with additional strategies.

- Interventions that indirectly assessed CT: The remaining 10 studies did not measure CT explicitly but through less specific instruments and indicators of change, primarily tests of knowledge and skills to detect mis- and disinformation within broader domains such as MIL or from implicit assumptions on the basis of theoretical frameworks and the design or content of interventions. This indirect approach highlights a degree of ambiguity in how CT was evaluated in every educational context and issue, resulting in heterogeneous conclusions and inferred improvements. Nevertheless, several common patterns suggest a positive impact on CT development, for example, more complex scrutiny of information (Al-Zou'bi 2022); improved media knowledge through MIL, particularly in content evaluation and verification (Murrock et al. 2018; Nazarnia et al. 2023; Şencan and Soydal 2023); reduced reliance on unreliable media and less dissemination of mis- and disinformation (Green et al. 2022; Lutzke et al. 2019; McLaughlin and McGill 2017); a greater capacity for credibility analysis and critical reflection (Cernicova-Buca and Ciurel 2022); and, finally, heightened engagement with relevant social and political issues (Yang et al. 2020). In terms of reported effectiveness, most of these interventions also had a positive impact. The potential of active learning methods (among active inoculation and serious games), explicit training and practical knowledge application in real situations seemed crucial for developing more prepared and presumably critical citizens in the face of mis- and disinformation.

Synthesis and general limitations of the analysed studies. Following the analysis above, when considering all interventions together, the majority ($n = 14$) reported improvements in CT following the application of diverse instructional approaches with different objectives and measurement strategies (see Table 3). This finding raises concerns about potential publication bias and the heterogeneity among the approaches used to foster CT in this context because they expose significant current inconsistencies in designs and measurement practices. Therefore, it is impossible to make a categorical statement about the effectiveness of these practices, the analysis of which is such a complex undertaking that it requires new research efforts.

Finally, it is worth mentioning that all the analysed studies highlighted concerns and suggested new lines of work for the future. The identified limitations emphasise challenges such as the representativeness of the samples, the robustness of the methodological design, the validation of measurement tools, and the influence of human and technological factors on the results and their generalisation. In response to these concerns, the various authors recommended conducting further research through experiments that employ diversified approaches, are more CT aware and include long-term follow-up. There is also a call to systematically integrate CT and MIL skills into curricula, design new training strategies and teaching tools on this issue and ensure rigorous evaluations to ascertain the effectiveness and transferability of interventions.

Risk of bias. Almost half of the sample ($n = 8$) was assessed using ROB-2.0. (Sterne et al. 2019) because it contained true experimental designs. Five studies had a low overall risk of bias with

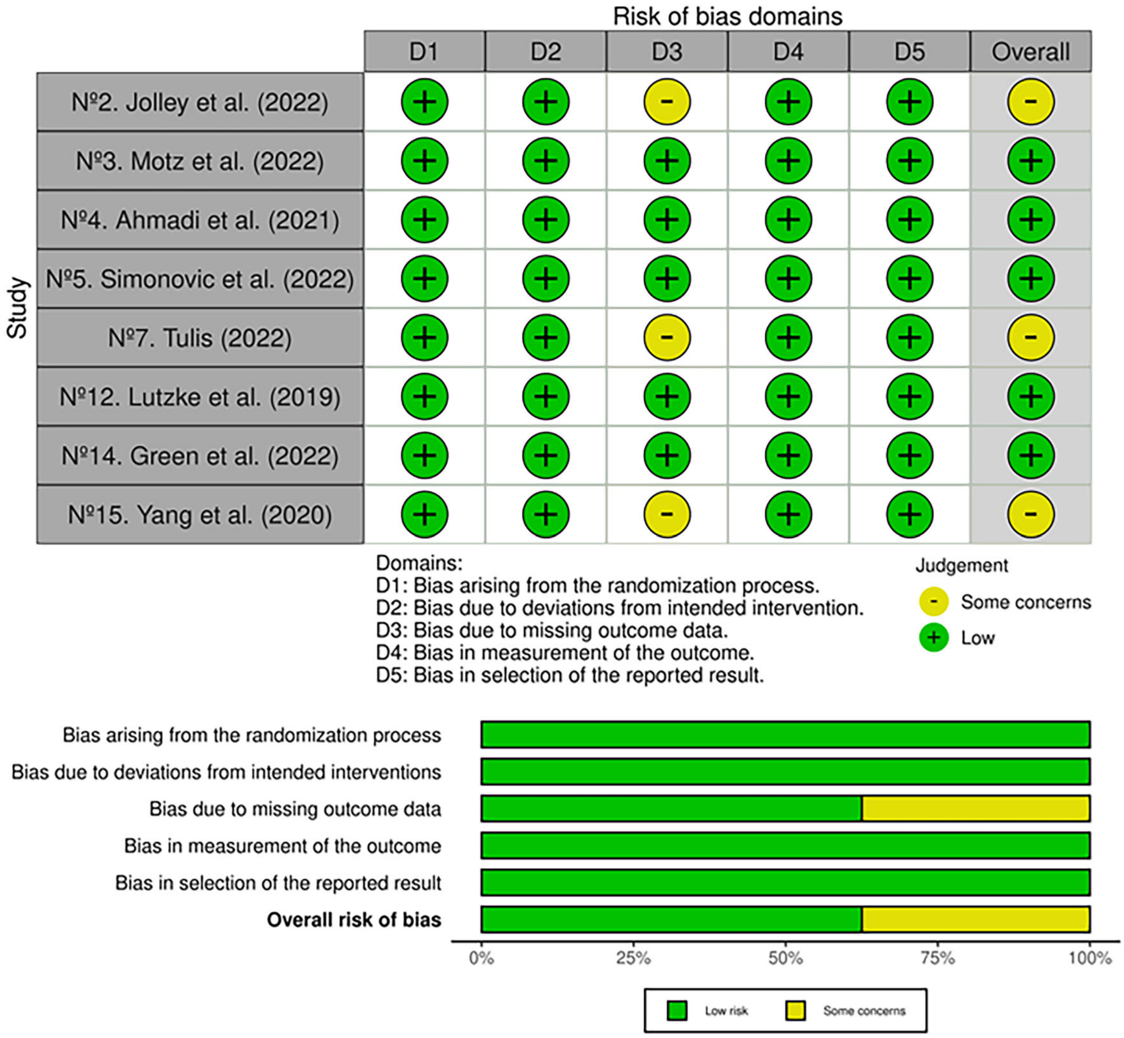

Fig. 3 Summary of the risk of bias assessment of randomised trials. Source: Authors.

good scores in all domains, whereas three presented a moderate level of bias because of concerns over missing data and their treatment (see Fig. 3).

The other half of the sample ($n = 9$) was assessed with ROBINS-I (Sterne et al. 2016) because it contained pre-experimental and quasi-experimental designs. None of these studies exhibited a low risk of bias, given the lack of information regarding the management of missing data and the assignment of moderate scores concerning the dangers of confounding bias (linked to potential factors that could influence the development and impact of the evaluated interventions), as well as other concerns regarding outcome measurement and sampling. With respect to the latter domain, four studies had a serious risk, as they contained high scores for potential bias in the non-random selection process of participants (see Fig. 4).

Although assessing study quality requires the consideration of multiple factors, analysing the risk of bias reveals significant strengths as well as limitations within the reviewed evidence base. The heterogeneity of the interventions indicates that some are methodologically more robust than others, with notable implications for both the conclusions of the findings of this SR and practical applications. In this sense, studies that demonstrated high methodological rigour ($n = 5$) provided stronger evidence to inform the development of educational strategies to address mis- and disinformation, whereas the majority raised concerns about the reliability of their results and their applicability ($n = 12$). The identification of five true experimental studies with a low risk of bias underscores the limited but promising feasibility of designing robust interventions in this field. Notably, three of these studies (Ahmadi et al. 2021; Motz et al. 2022; Simonovic et al. 2022) explicitly addressed CT in the context of disinformation, offering valuable insights for effective strategies. The other two studies, with moderate risks of bias, warn of the need to improve data collection and management procedures, as this may reduce the statistical power and bias the interpretation of the results. In contrast, the absence of quasi-experimental or pre-experimental studies with a low risk of bias emphasises the critical need for more rigorous research designs. Thus, particular attention must be paid to random participant selection and the provision of comprehensive information to minimise limitations in establishing causal relationships.

Discussion and conclusion

This SR aimed to collect and systematise the key elements of training actions designed to improve CT in the face of mis- and disinformation, while evaluating their results to determine the effectiveness of such interventions. The findings provide valuable insights into how CT is being addressed in this context, outlining the current state of research, including promising strategies, underlying challenges, and potential opportunities for future efforts.

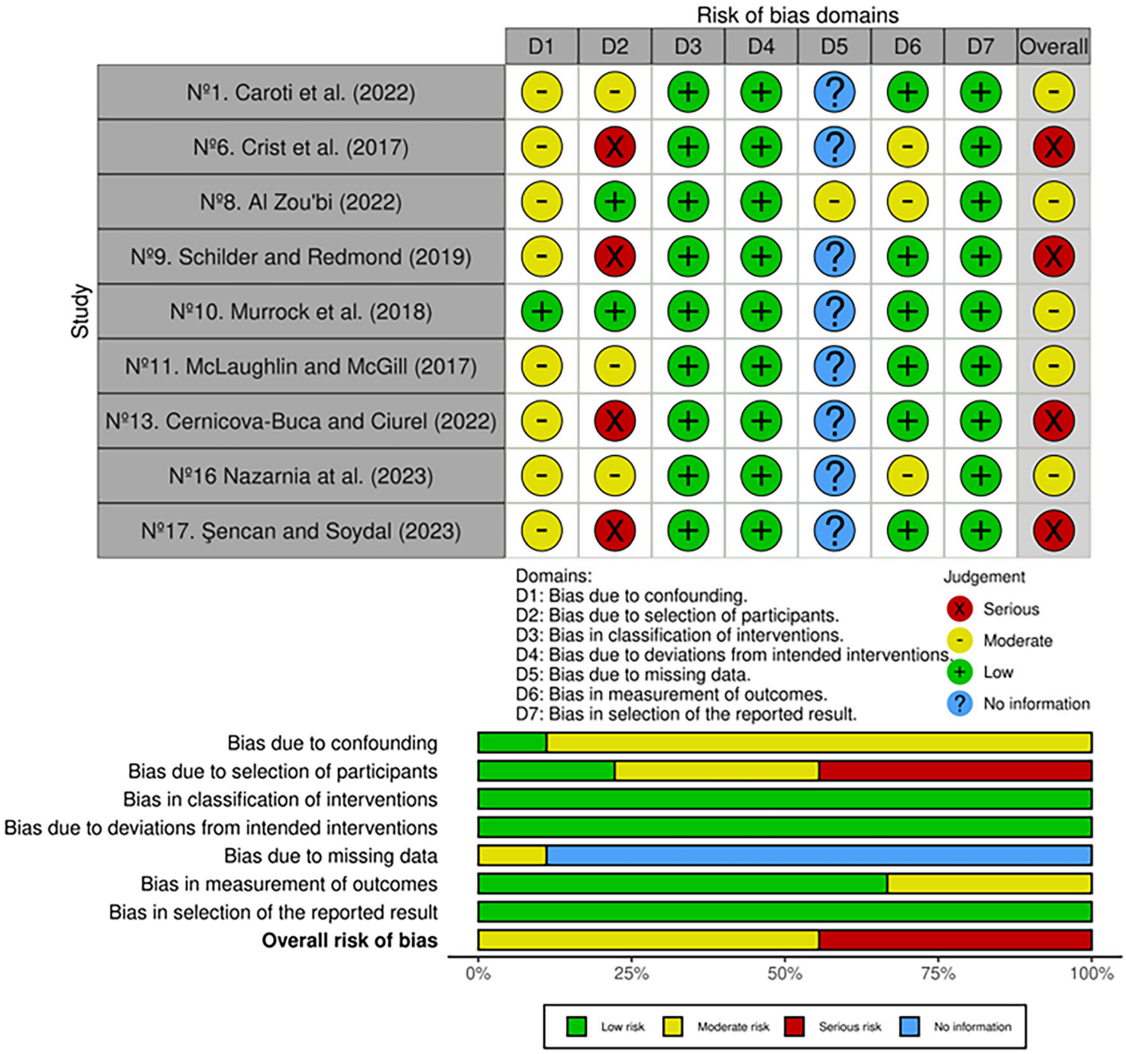

Fig. 4 Summary of the risk of bias assessment of non-randomised studies of interventions. Source: Authors.

The following section provides a synthesis of the evidence related to the stated objectives, followed by a discussion of the study's overall conclusions, organised around several key dimensions in broader theoretical and practical debates.

Responses to the research objectives: main findings

Specific objective 1: identifying and compiling evidence. This review collected a relatively small but significant set of empirical studies ($n = 17$). Of these, only seven met the eligibility criteria with sufficient rigour, placing CT development explicitly at the core of their objectives and employing dedicated assessment instruments. Despite the limited sample size, the total collection of entries highlights a clear upward trend in scientific production since 2020, with a notable presence in high-impact journals. Furthermore, the geographical diversity of the authorships, along with their interdisciplinary approaches and range of methodologies, suggests that CT is an increasingly global concern.

Specific objective 2: systematising key elements of the training actions. The reviewed studies revealed a heterogeneous methodological and pedagogical landscape. Only four interventions prioritised the deliberate teaching of CT through "infusion"-based strategies, which align with documented trends about CT instruction (Abrami et al. 2008; Behar-Horenstein and Niu 2011; Lombardi et al. 2021; Tiruneh et al. 2014). The remaining interventions adopted a more cross-cutting approach, integrating CT alongside other complementary goals or within broader frameworks, often linked to mis- and disinformation management, such as MIL.

Despite these differences, several common elements were identified: most interventions took place in the context of Higher Education (HE), utilised active learning methodologies, incorporated multiple resources (primarily digital), and fostered collaborative and cooperative environments. However, the lack of detail regarding the duration of instruction, the role and qualifications of trainers, or the contextual conditions has complicated accurate assessment of the interventions' impact. This reflects the contradictions and informational gaps noted in the literature (Cáceres et al. 2020; El Soufi and See 2019), highlighting the need for greater transparency and standardisation in study designs to enhance replicability and support cumulative knowledge building.

Specific objective 3: evaluating the results of the training actions. Regarding impact, nearly all the studies reported at least short-term improvements in participants' CT. However, interpreting these results requires caution for several reasons.

Firstly, there is a potential risk of publication bias, which may have favoured the dissemination of positive findings over negative

Table 4 Summary of deviations from the protocol.

Deviations from the protocol - PRISMA 2020 statement (Page et al. 2021)

Protocol element	***Original plan***	***Final implementation***	***Justification***	***Manuscript section***
Expected outcomes (PICOS)	Inclusion of studies reporting direct, empirically measured CT improvements	Also included are studies offering strong inferential links to CT development, without direct measurement	Scarcity of direct empirical evidence, supported by methodological literature on reviews in emerging fields.	Section "Search strategy and study selection process".
Type of intervention (PICOS)	Inclusion of studies that develop training actions, preferably through instructional interventions.	Also included psychological interventions explicitly targeting CT against mis- and disinformation	Better reflects the actual diversity of interventions in the field.	Section "Eligibility criteria".
Data synthesis	Quantitative synthesis (meta-analysis)	No meta-analysis conducted due to high heterogeneity	Methodological and outcome diversity prevented a rigorous quantitative synthesis.	Section "Data collection and synthesis" Section "Discussion and conclusion".
Quality appraisal (GRADE)	Use of the GRADE framework.	Discarded in favour of individual risk-of-bias assessment.	GRADE was unsuitable given the methodological variety; space limitations also influenced the decision.	Section "Risk of bias".

Source: Authors.

ones (Franco et al. 2014). Moreover, considering the tools used, a methodological quality assessment of the individual studies identified only five with low risk of bias. In contrast, the remainder presented moderate to severe methodological shortcomings, raising concerns about the reliability and generalisability of their conclusions.

Secondly, there are notable inconsistencies in the conceptualisation, operationalisation, and measurement of CT. In many cases, it was assumed that participation in information verification-oriented or MIL-focused interventions would lead to improved CT, without directly assessing this outcome. Consequently, the effects of most interventions have not been measured based on explicit CT development but rather inferred from an assumed improvement in participants' CT capabilities post-intervention. Furthermore, within indirectly assessed CT, usage spans a continuum from substantive integration to more tangential invocation, for example, Green et al. (2022) and, to a lesser extent, Al-Zou'bi (2022). These focus and labelling decisions, rather than providing certainty, underscore the challenges of measuring CT accurately and the need to align frameworks, design and outcome indicators, to avoid a merely nominal use of the term 'critical thinking'.

Finally, the wide variety of measurement tools and strategies employed further complicates the comparison of results. Of all the studies, only a few used validated instruments specifically designed to assess CT, and even among these, there were notable differences in the subcomponents of CT being measured. This lack of standardisation, coupled with other reported limitations—such as the limited representativeness of samples, the short duration of interventions, and the absence of long-term follow-up—has restricted the generalisability of findings and poses significant challenges for future research.

General conclusions from a critical perspective

CT in the face of mis- and disinformation: a necessary but controversial imperative. Although interest in CT has grown considerably, there remains a persistent lack of conceptual clarity and consensus regarding its development. While all the interventions analysed aimed to enhance CT, only a few explicitly defined it as a central objective. This ambiguity is not trivial, as it reflects a deeper tension between the growing normative demand to foster CT (change to Scott 2017)—particularly in response to contemporary informational threats (Wikforss et al. 2019)—and the ongoing challenge of translating this demand into effective practical applications (Dwyer 2023). In this context, functional definitions of CT have gained increasing importance (Bergstrom & West 2021; Grizzle et al. 2021; Schmaltz et al. 2017; Wilson 2011). This operational shift may seem reductionist, but it is a strategic move, allowing CT to be integrated into a wide range of educational actions, including AMI programmes (Alcolea-Díaz et al. 2020; Andersson 2021; López-González et al. 2023).

From this perspective, CT is a fundamental means of fostering citizens' epistemic agency (Brisola and Doyle 2019; Higdon, 2020; Molerov et al. 2020). However, as previously noted, this convergence can risk a counterproductive simplification (Boyd 2017, 2018; Nygren and Guath 2019; Potter 2018). In response to this challenge, several scholars (Bennett & Livingston (2018); (Phippen 2021) have advocated for a more CT-centric approach to AMI, underscoring the need for a coherent articulation between both competencies. This approach acknowledges that, while AMI and CT share a commitment to informational autonomy and reflective judgment, CT introduces a deeper, almost philosophical dimension: an existential orientation towards knowledge and truth-seeking (Gallego Torres 2023; Jiménez-Aleixandre and Puig 2022).

Unfortunately, this SR proves that this symbiosis is still far from fully realised. Only a minority of the interventions identified CT as a distinct competency, while most inferred critical improvements without clearly detailing the mechanisms through which these changes were generated and assessed. Therefore, many initiatives—beyond those examined in this review—may be operating under implicit assumptions about CT enhancement, without a clearly defined theoretical foundation.

Teaching CT: from active learning to reflective engagement. Although the studies analysed included demographically diverse samples, most interventions were conducted within university settings. This finding reflects a trend as widespread as it is contested in empirical research, particularly in education and the social sciences, where students are frequently selected as convenience samples given their accessibility to researchers. Consequently, this predominance constitutes a limitation that should be progressively addressed in future research, as it restricts the generalisability of findings beyond this specific population.

Nonetheless, the focus on university students is not merely a matter of methodological convenience; it also reflects a genuine and well-supported concern in the literature regarding the importance of fostering CT during a formative stage in the development of young people's personal and social identities (Herrero-Diz et al. 2023; Phippen et al. 2021). Large-scale data confirm a legitimate concern about this age group's particular mis and disinformation exposure. According to the latest Eurobarometer Youth Survey (European Parliament, EP 2025), over 76% of respondents reported encountering disinformation "occasionally", "often", or "very often". While many claimed to feel capable of identifying such content, broader research reveals important discrepancies. For instance, recent findings highlight the vulnerability of university students due to their limited training and uncertainty about how to respond effectively (Del Moral Pérez et al. 2025). Similarly, a scoping review identifies young people as at risk because they tend to overestimate their evaluative and detection skills, exacerbating the challenge (Kops et al. 2025). This scenario especially concerns today's global context, where the World Economic Forum (2025) has identified disinformation as the most significant risk to modern societies. As such, focusing on this population remains especially pertinent in an era of post-truth, where the intensive use of digital technologies and the rapid proliferation of generative AI amplify young people's exposure to misleading or manipulative content, further blurring the boundaries between truth and falsehood (Corbu et al. 2021; Del Moral Pérez et al. 2025; Deroncele-Acosta et al. 2020; Falloon 2024; Grizzle et al. 2021).

Within this context, HE plays a critical role in preparing critically engaged citizens before they enter the labour market and participating meaningfully in public life (Altuve 2010; Bezanilla-Albisua et al. 2018; Van Damme and Zahner 2022). Active learning methodologies emerge as an effective pedagogical response, as evidenced by the parallels between our findings and previous studies (Abrami et al. 2015; Bezanilla-Albisua et al. 2019; Dumitru et al. 2018). These constructivist approaches promote student engagement, the dialogic construction of knowledge, and the confrontation of ideas within realistic and complex scenarios (Anggraeni et al. 2023; Collazos Alarcón et al. 2020; Jaramillo Rivadeneira et al. 2023; Kusumoto 2018; Nelson and Crow 2014). Nevertheless, this orientation does not always result in deep reflective engagement. Indeed, this SR identified limited attention to components oriented towards metacognition. For example, aspects such as the self-regulation of thinking and epistemic humility—both increasingly recognised as essential to counter mis- and disinformation (Andersson 2021; Butler 2024)—were scarcely addressed. This is concerning, as fostering critical awareness of personal biases and reasoning patterns is crucial to avoid excessive scepticism and the counterproductive effects of overcorrection of false or misleading content (Li 2023; Nygren and Guath 2019).

In response to this gap, there is an opportunity to embrace more transformative educational approaches, such as that inspired by Freirean critical pedagogy (Freire 1970), which not only advocates teaching CT but also emphasises understanding why this is important: to resist manipulation, promote social responsibility, and uphold democratic discourse (Chibás Ortiz and Novomisky 2023). Therefore, for HE to fulfil its role as a genuine agent of change, it is essential to advance towards authentic reflective engagement that integrates the epistemic, ethical, and civic dimensions of CT together (Chibás Ortiz and Novomisky 2023; Jiménez-Aleixandre and Puig 2022; Osuna-Acedo and Feltrero 2023).

Assessing CT: a major methodological challenge. Ultimately, one of the critical challenges identified in this SR is the absence of methodological consistency in defining and applying strategies for assessing CT. This inconsistency not only limits the ability of researchers to identify effective patterns and generalise scientific findings but also reduces the capacity of educators to design and implement formative interventions with demonstrable impact. The OECD has highlighted significant disparities in how CT is addressed internationally, creating fertile ground for divergent interpretations and inconsistent results (Van Damme and Zahner 2022).

This issue is largely due to conceptual differences regarding the very construct of CT, which, already complex by nature, becomes even more challenging to assess comprehensively, as it involves both cognitive skills and emotional dispositions (Nieto and Sainz 2010; Ossa-Cornejo et al. 2017). Indeed, recent studies emphasise that assessment should be a central concern in advancing CT development, with future efforts directed not only towards the technical aspects of psychometric validation and the tendency to measure success based solely on correct answers, but also towards evaluating the quality of reasoning (Butler 2024; Hart et al. 2021)

Main contribution of the SR. To conclude, this SR extends and updates the path established by previous reviews, such as the one by Machete and Turpin (2020), contributing to a deeper understanding of how to foster CT to counter mis- and disinformation. It identifies persistent conceptual and methodological gaps, underscores the need for integrated frameworks in which CT is explicitly, intentionally, and systematically developed in its full scope, and highlights the importance of adopting aligned assessment practices to ensure reliable and comparable results. In summary, its main contribution lies in recognising these pressing challenges while encouraging further progress in addressing them. Determining the effectiveness of interventions aimed at this goal remains an open and necessary area for future research.

Limitations of the SR. Some evident limitations include the scarcity of studies meeting the eligibility criteria, the heterogeneity of the interventions analysed, and the lack of robust quantitative data. Additional challenges stem from potential inconsistencies in the conceptualisation and evaluation of key constructs—namely, CT and the threat of mis- and disinformation—which, together with factors such as the specific search and selection strategies employed and the time frame of this review, may have led to the exclusion of recent and relevant interventions conducted in similar contexts.

In particular, the decision to focus exclusively on studies that explicitly refer to CT was methodologically justified and aligned with the scope of this SR. However, this criterion may have limited the inclusion of interventions that, while less precise in their terminology, could nonetheless be relevant to the object of study. Exploring alternative labels and approaches to the CT construct would represent a complementary avenue for further understanding how it is operationalised in practice. This remains a complex challenge, but also a promising future research direction.

A meta-analysis could have provided greater clarity and helped address the uncertainties within this field; however, such an approach was not feasible due to the abovementioned limitations.

Future implications of the SR. The still incipient nature of the available research corpus highlights the need to expand the empirical base through more rigorous and diverse studies, contributing to the consolidation of the field and informing evidence-based interventions.

Future studies should examine the long-term impact of CT instruction across diverse populations, socio-cultural contexts and educational levels, including its applicability in informal learning environments. Such approaches should conceive CT not merely as a learning goal, but as a fundamental pathway towards cultivating a better-informed and more resilient society. This requires placing CT at the heart of educational efforts and evaluating it systematically and intentionally, rather than presuming that any intervention will inherently lead to meaningful outcomes.

In parallel, it would be valuable to complement this SR with further literature-based inquiries, such as future reviews or meta-analyses that both build on the present work and, at the same time, open alternative avenues. One promising direction would be to update and deepen the analysis of available evidence within frameworks that explicitly define CT, such as Facione's (1990) taxonomy, to refine our understanding of its constituent skills and dispositions. Initiatives like this would help broaden our insight into the diverse ways CT can be fostered and allow for a more accurate synthesis of new research findings.

Finally, all these CT-centred proposals should be integrated into a broader, more holistic framework of solutions to effectively counter mis- and disinformation (Herrero-Diz et al. 2023; Jackson 2019; Manassero-Mas et al. 2022). Only by adopting this comprehensive perspective can the theoretical foundations advocated in this study acquire genuine practical significance. This view aligns with discourses of growing relevance, such as that of McIntyre (2019, 2021, 2023), who advocates for positioning CT as a vital attitude that fosters reflection and serves as a powerful exercise of intellectual freedom in the challenging times we face.

Differences between the protocol and the SR

This SR presents some differences from the original protocol study (Marcos-Vílchez et al. 2024). All modifications were introduced during the work process and were agreed upon by the research team, following the methodological guidelines proposed by the PRISMA 2020 Statement (Page et al. 2021) and other recommendations of reference frameworks for literature reviews, especially in complex fields (Booth et al. 2021; Higgins et al. 2024).

The main deviations and their justifications are detailed below, together with the relevant sections in the manuscript where they are discussed (Table 4):

Study selection process. Due to the limited availability of studies that strictly met the initial eligibility criteria, particularly regarding the expected outcome of directly and specifically measured CT improvements, we introduced a controlled adjustment: the inclusion of studies that, while not reporting direct empirical results, provided a well-founded rationale linking the intervention to CT development. This decision is justified by methodological literature and detailed in Section "Search strategy and study selection process". Search strategy and study selection, which allowed us to broaden the analytical base of the review and distinguish between interventions offering direct measurements and those presenting inferential or indirect assessments, without compromising the overall methodological rigour.

In addition, the scope of included interventions was expanded to cover both instructional and psychological approaches, but with the training objective of improving CT against mis- and disinformation.

Data synthesis. The quantitative synthesis originally planned in the protocol was not conducted due to the high heterogeneity of the included interventions, the disparity in the variables of interest, and the difficulty in presenting refined results. Consequently, the fourth additional research objective, which aimed at conducting a meta-analysis to assess the overall effect of training actions designed to improve CT in the face of mis- and disinformation, could not be fulfilled (Marcos-Vílchez et al. 2024).

Classification and evaluation of the quality of the evidence. The GRADE system of evidence assessment was discarded in favour of prioritising the analysis of study biases due to both practical constraints (e.g., length limitations) and the methodological diversity of the included studies.

All these changes were made to enhance the review's rigour, analytical depth, clarity, and relevance. Each modification was documented and approved by the authors involved.

Data availability

Additional information that complements the set of results of this SR is provided in the Appendices. Data sharing does not apply to this article, as no datasets were generated or analysed during the current study.

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Notes

1 A glossary of abbreviations and acronyms is included in the Appendices (see Appendix I) to facilitate easier comprehension and enhance the readability of this article.

2 The information about the research objectives and the explanation of the method closely align with the details provided in the SR protocol article. The reason is that the current study is based on the protocol paper published by the authors and cited in this manuscript to avoid plagiarism or self-plagiarism.

3 Although the PICO strategy developed for the original protocol focused on instructional interventions, the corresponding eligibility criterion (type of intervention) was subsequently broadened to include psychological interventions explicitly aimed at fostering CT in response to mis- and disinformation. This decision followed a deeper engagement with the theoretical framework of the SR, which clarified that both instructional and psychological approaches can be meaningfully operationalised as formative actions. The adjustment reflects the practical convergence of both types of interventions within this field and is documented in Section "References". Differences between the protocol and the SR.

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Author contributions

JMMV: Guarantor author. This author has led the study, participating in all the processes: conceptualisation, methodology, literature review, selection and analysis of the studies found, extraction of information and synthesis of the results. This author has written the manuscript and is the corresponding author of the submission. JAMV: This author coordinated the development of the study, paying attention to thematic coherence and the definition of the research objectives. This author also participated in the second screening (peer review of articles) to verify the results of the full-text studies and finally reviewed the entire manuscript. AAC: This author made a substantial contribution to the first peer review of the initial sample of entries, reviewing the titles and abstracts of the articles. This author also participated in the final data extraction, the risk of the bias assessment process. MSM: This author coordinated the development of the study, being in charge of the methodological design and the control of the fulfilment of all transparency and quality requirements for the conduct of the SR, intervening to reach agreements after the anonymous peer reviews. In addition, this author conducted a thorough review of the manuscript before submission. All authors reviewed the results and approved the final version of the manuscript.

Competing interests

JAMV was a member of the Editorial Board of this journal at the time of acceptance for publication. The manuscript was assessed in line with the journal's standard editorial processes, including its policy on competing interests. The authors declare no competing financial interests.

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